Predictive Modeling Using Machine Learning - Project Code

Sample internship project using Random Forest Classification.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("data.csv")

# Example: Last column is target variable
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))

# Feature Importance
importance = model.feature_importances_

plt.figure(figsize=(8,5))
plt.bar(range(len(importance)), importance)
plt.title("Feature Importance")
plt.xlabel("Features")
plt.ylabel("Importance")
plt.show()

print("Predictive Modeling Project Completed")
```